Building an AI-Powered Quran Project: A Feasibility Guide and Scoped FYP Ideas for a 4-Person AI Student Team

TL;DR

The single most buildable, academically defensible FYP for your team is a *scoped* Quranic recitation error-detection + memorization assistant for a limited set of surahs (e.g., Juz Amma), built on the open-source `tarteel-ai/whisper-base-ar-quran` model (eval WER dropped from 13.38% to 5.75% over 5,000 steps) plus a fuzzy text-alignment layer — this reuses proven building blocks while letting you contribute something new (e.g., a specific tajweed rule, non-native-speaker feedback, or an under-served language). Do not try to clone all of Tarteel; scope tightly.
The whole AI+Quran space is mature at the top (Tarteel self-reports "15+ million" users and runs a proprietary NeMo-based real-time engine) but shallow in the open middle, so the realistic student niche is a *narrow, well-evaluated* system built from open datasets (Tarteel/EveryAyah audio, QDAT tajweed labels, Quranic Arabic Corpus, Tanzil text) and open models (Whisper fine-tunes, wav2vec2, RAG over tafsir).
The two biggest risks are religious/scholarly accuracy (hallucinating chatbots, wrong tajweed feedback) and data scarcity for tajweed-labeled audio — plan for a scholar/qari reviewer, a tightly-scoped rule set, and honest evaluation metrics, and your project becomes legitimate research rather than a fragile demo.

Key Findings

Existing apps split into two clusters. (a) *Recitation/ASR apps* (Tarteel, TajweedMate, HifzAI, Quoranize, quran.ai) that listen to your voice and give feedback; and (b) *Q&A/chatbot apps* (the many "Noor AI"/"NoorAI"/NurAI variants, Quran.ai by Boulevard, MyQuran) that answer Islamic questions. The first cluster is technically hard and mostly proprietary; the second is easy to prototype but hard to make *trustworthy*.
Tarteel is the benchmark and it is largely closed. It open-sourced a 25,000-clip / 67.39-hour crowd-sourced dataset (NeurIPS 2021 Datasets & Benchmarks Track) and Whisper fine-tunes on Hugging Face, but its production real-time mistake-detection engine is proprietary (NeMo-based, sub-200ms latency). This is exactly why a student project should *not* aim to beat Tarteel on general recitation, but to do something narrower and better-evaluated.
The open building blocks are excellent and free. Text (Tanzil, Al Quran Cloud, Quran.com/Quran.Foundation API, fawazahmed0's 400+ translations), audio (EveryAyah, QuranicAudio, MP3Quran, OpenSLR-132), morphology (Quranic Arabic Corpus / JQuranTree / QuranTree.jl), tafsir (quran-tafseer.com API, Quran.com), hadith (sunnah.com API, AhmedBaset/hadith-json with 50,884 hadiths from 17 books), tajweed-labeled audio (QDAT, ~1,505 clips, 3 rules), and ASR models (`tarteel-ai/whisper-base-ar-quran`, wav2vec2/Conformer-CTC Arabic fine-tunes).

Academic precedent exists for every core feature, which is good for your literature review: Quranic ASR (DeepSpeech, Whisper, Conformer), tajweed classification (MFCC+LSTM at 96%/95%/96% on QDAT), semantic search (QSST, QurSim, arXiv 2311.05120), Quranic OCR (CNN+RNN on Mushaf al-Madinah, QTID with 309,720 word images), and RAG Q&A (Aftina fatwa RAG, IslamicLLM surveys).
The differentiator that turns a clone into a project is almost always one of: a *specific under-served rule/skill* (e.g., one tajweed rule with detailed diagnosis), a *specific under-served user group* (children, non-native/non-Arabic speakers), a *specific language* (Urdu/Sindhi/Pashto feedback for Pakistani learners), *rigorous evaluation on a released benchmark*, or *grounded, citation-first RAG with scholarly review*.

Details

1. Landscape of existing AI + Quran apps and tools

Tarteel AI (market leader). Uses speech recognition + NLP for real-time, word-level recitation tracking: it transcribes the recitation and does *text-based fuzzy matching* against the canonical Quran text (edit-distance alignment) rather than heavier waveform methods like Dynamic Time Warping. Features: real-time mistake detection (missed/incorrect/skipped words, tashkeel issues), "AI Voice Search" ("Shazam for the Quran"), Hidden Verses / blind-recitation mode, memorization goals, streaks/heatmaps, historical mistakes tracking and adaptive revision. Maturity: very high; its own App Store/Google Play listing markets it as "THE #1 AI QURAN COMPANION" with "15+ million" users (a self-reported marketing figure, not independently verified). Openness: **partly open** — the *Tarteel Dataset* (25,000 clips, 67.39 hours, 1,200+ contributors over six months, released CC BY-NC-ND 4.0 via the NeurIPS 2021 Datasets & Benchmarks Track, Khan, Abou Allaban, Moussa & Abid) is published, and Whisper fine-tunes (`whisper-base-ar-quran`, `whisper-tiny-ar-quran`) and `everyayah` data are on Hugging Face; the production engine (NeMo-based, real-time) and larger private datasets are closed.

TajweedMate, HifzAI, Quoranize, gethifz, quran.ai — a wave of newer AI recitation/hifz apps. TajweedMate records recitation and gives AI feedback on pronunciation/tajweed and articulation points (makharij). HifzAI and Quoranize combine spaced-repetition (SM-2-style) memorization with AI voice-recitation checking. quran.ai is building "infrastructure for AI" — an MCP server exposing Quran text, 50+ translations, tafsir, semantic search, word morphology, plus "Recite" (word-by-word tilawah tracking). Maturity: medium; mostly closed-source; techniques are the same family as Tarteel (ASR + alignment, SRS scheduling). Tajweedmate ↗
Quran ↗

Noor AI / NoorAI / NurAI (multiple different apps) — a crowded set of Islamic Q&A chatbots, not one product. Examples: Noor AI (rebounddev), NoorAI (Gowalk, "studied with imams"), NurAI (Zetrix, Malaysian-fatwa focused, cites JAKIM/national fatwa council), Noor AI (noorai.online, Bangla+English over Quran/Hadith). These are LLM chatbots (often RAG or prompt-engineered over Islamic sources) answering questions on fiqh, tafsir, duas. Maturity: low-to-medium; almost all closed; quality/authenticity varies widely.

Quran.com / Quran.Foundation — the reference reading platform and, increasingly, an AI-infrastructure provider: open APIs, an MCP server for "semantic search and seamless content retrieval from Quran.com's verified text, translations, tafsir, and word-for-word morphology," plus connected apps like Tazkiya (verse-by-

verse, source-cited Q&A) and an AI-assisted verse-alignment video subtitler. Openness: **high** — much is open-source on GitHub. Quran.com ↗

Ayat (KSU) & King Fahd Complex (KFGQPC) apps — authoritative but *not AI*. Ayat (King Saud University's Electronic Moshaf) integrates Quran images (KFGQPC), text/translations (Tanzil), tafsir (mosshaf.com), and audio (Verse-by-Verse) — a great data-integration model to emulate. KFGQPC's own apps are high-resolution mushaf readers with search, indexes, and verse-tracked audio; no meaningful AI. Quran ↗

Quranic Arabic Corpus (corpus.quran.com) — University of Leeds annotated linguistic resource: 77,430 words with part-of-speech, morphology, a syntactic treebank (i'rab dependency grammar), and a semantic ontology. Accessed via JQuranTree (Java), QuranTree.jl (Julia), or the mustafa0x/quran-morphology GitHub fork. GNU-licensed download available. This is the backbone for any Quranic NLP/grammar/semantic-search project. Wikipedia ↗
GitHub ↗

Tafsir/Q&A AI tools & research — quran-tafseer.com (free tafsir API, multiple classical tafsirs), Quran-Hadith RAG projects on GitHub (e.g., hammadali1805 using ChromaDB + all-MiniLM-L6-v2 + Gemini query expansion; Shariar-Mozumder's Llama-3.2-1B RAG; Shaheer66's Islam-GPT), the Itqan computational Quran-Hadith concordance (FAISS, multilingual-e5, Qwen2.5-1.5B RAG), and academic fatwa-RAG systems (Aftina, Turkish religious-QA arXiv 2401.15378).

2. Technical building blocks (feasibility-level detail)

(a) Speech-to-text / ASR for Quranic recitation. The dominant approach is fine-tuning a pretrained model. Off-the-shelf open options: `tarteel-ai/whisper-base-ar-quran` (apache-2.0, ~292 MB; eval WER fell from 13.38% to 5.75% over 5,000 training steps), `whisper-tiny-ar-quran`, `OdyAsh/faster-whisper-base-ar-quran` (float16, 141 MB), LoRA fine-tunes (KheemP, WER ≈5.98% with diacritics), and Conformer-CTC Arabic models. Data: EveryAyah (44 renowned professional *male* reciters, all 114 surahs, verse-level with full diacritics, studio-quality — ~1,310 hours in one comparative study, per arXiv 2606.19747), the Tarteel crowd-sourced set (25k clips / 67h, real phone recordings, variable quality), OpenSLR-132, Kaggle "Quran ayat speech-to-text" (32 reciters), MohamedRashad/Quran-Recitations. Feasibility: **using** these models is easy (a Colab notebook + `transformers` + `librosa`); **fine-tuning** Whisper-small on a single GPU is a standard, well-documented FYP-scale task. Key nuance: diacritics matter, and professional-male-only training data generalizes poorly to women's/children's/non-native voices — a documented gap you can target. Hugging Face ↗
Hugging Face ↗

(b) Tajweed rule-violation detection. Two families. *Rule-based / signal-processing*: extract features (MFCC, LPC, WPD, pitch/duration via librosa or Parselmouth/Praat) and check against expert reference (e.g., Madd elongation = duration; Ghunnah = nasalization; Qalqalah = echo) — interpretable but brittle. *ML-based*: classify mel-spectrograms/MFCCs with CNN/CRNN/LSTM/EfficientNet. The public **QDAT dataset** (~1,505 clips from Surah Al-Ma'idah v.109, binary labels for 3 rules — Al Mad/separate stretching, Ghunnah/tight noon,

Ikhfaa/hide) is the standard benchmark. Best published results: Al Harere & Al Jallad (arXiv 2305.06429, May 2023) report "96%, 95%, and 96% for the three rules (Separate stretching, Tight Noon, and Hide)" with MFCC+LSTM; Shaiakhmetov et al. (arXiv 2503.23470, Mar 2025) report "95.35% for separate stretching (Al Mad), 99.34% for tight noon (Ghunnah), and 97.01% for hide (Ikhfaa)… average accuracy was 97.23%, which is higher than the benchmark of 95.33% set by Harere and Jallad" using EfficientNet-B0 + Squeeze-and-Excitation. Frontier: the Ain Shams University group (Abdelfattah, Khalil, Abbas; arXiv 2509.00094, listed at ICML 2026) introduced a **Quran Phonetic Script (QPS)** — 43 phoneme symbols plus a Sifa (articulation-attribute) level of ten classical articulation attributes = 11 parallel CTC heads on a Wav2Vec2-BERT model, trained on "848 hours of audio (286K annotated utterances)" via a 98%-automated pipeline ("Tasmeea" edit-distance verification; a fine-tuned wav2vec2-BERT waqf segmenter at F1 ≈0.995), reporting "0.16% average Phoneme Error Rate (PER) on the testset" and a `qdat_bench` benchmark of 159 real-error samples; code/data released. Feasibility: **classifying a small fixed rule-set on QDAT is a realistic FYP**; full phoneme-level diagnosis like the QPS system is beyond a first FYP (needs large annotated data + heavy compute).

(c) Quranic Q&A / tafsir chatbot via RAG. Standard, very student-friendly stack: chunk tafsir/translation/hadith corpora → embed (all-MiniLM-L6-v2, multilingual-e5, or Arabic-native embeddings) → store in a vector DB (Chroma, FAISS, Qdrant, Pinecone) → retrieve top-k → feed to an LLM (open Llama/Qwen/Mistral or an API) with a citation-first prompt. Add a re-ranker and semantic-similarity filtering to cut hallucinations (as in the Aftina fatwa-RAG work). Feasibility: **a working prototype is a weekend; a *trustworthy* one is the whole project.** The academic value is entirely in grounding, citation accuracy, refusal behavior, and evaluation — not the plumbing.

(d) Semantic search over verses/translations. Embed verses + translations + tafsir and rank by cosine similarity. Precedent: QSST (Word2Vec CBOW over annotated Quran), QurSim / multilingual-QurSim (verse-relatedness pairs), and arXiv 2311.05120 (semantic search over 30+ tafsirs, cosine sim up to 0.97). Feasibility: **easy-to-medium**; a strong standalone project if paired with good evaluation and a real UI.

(e) Open datasets & APIs (all free). Text: **Tanzil** (Uthmani/simple, CC-BY-verbatim), **Al Quran Cloud** (`api.alquran.cloud`, 50+ translations + tafsir + audio CDN, no key), **Quran.com / Quran.Foundation** (OAuth, MCP, morphology), **fawazahmed0/quran-api** (400+ translations, 90+ languages). Audio: **EveryAyah**, **QuranicAudio**, **MP3Quran**, OpenSLR-132. Tafsir: **quran-tafseer.com** API, Quran.com. Hadith: **sunnah.com** API (key by request), **AhmedBaset/hadith-json** (50,884 hadiths, 17 books incl. the nine books, Arabic+English+grading), meeAtif/hadith_datasets. Morphology: **Quranic Arabic Corpus** (v0.4, GNU). Tajweed audio: **QDAT**. This abundance means **your team should spend ~zero time on data plumbing and all your time on the AI contribution.**

(f) OCR (bonus building block). Quranic OCR is a solved-ish research niche: CNN+RNN (LSTM/GRU) on the Mushaf al-Madinah benchmark (604 pages / 8,927 text-line images) and **QTID** (309,720 word images, 2.49M characters, first Arabic dataset with diacritics). Useful if you want a "point your camera at a mushaf" feature, but Arabic diacritic OCR is genuinely hard. Academia.edu ↗

3. Ranked FYP project ideas (easiest/most buildable → more ambitious)

1. Grounded, citation-first Quran & tafsir Q&A assistant (RAG) — EASIEST, high polish.

Description: A chatbot/search tool that answers questions **only** from a fixed, vetted corpus (a chosen translation + 2–3 classical tafsirs + optionally sahih hadith), always returning the exact verse/tafsir citation, and *refusing* or deferring to a scholar when confidence is low or the question is a fatwa.
Core AI: RAG (embeddings + vector DB + LLM), re-ranking, hallucination/abstention control.
Data/APIs: Tanzil / Al Quran Cloud text, quran-tafseer.com, sunnah.com or hadith-json, Chroma/FAISS/Qdrant.
Difficulty: Low-medium. *Differentiator vs. the "Noor AI" crowd:* rigorous **citation-accuracy and hallucination evaluation**, explicit madhhab/source transparency, refusal behavior, and a scholar-reviewed test set — the very things existing chatbots skip.

2. Semantic verse search for a specific theme/language — EASY.

Description: "Find every ayah about X" (patience, charity, parents…) via meaning, not keywords — ideally in Urdu/Sindhi/English for local users.
Core AI: Multilingual sentence embeddings + cosine similarity; optional cross-encoder re-ranker.
Data: Tanzil + fawazahmed0 translations + tafsir; benchmark against QurSim.
Difficulty: Low-medium. *Differentiator:* build and **release a labeled evaluation set** for your target language; compare embedding models (a clean empirical contribution).

3. Recitation follow-along + memorization (hifz) assistant for a limited scope (e.g., Juz Amma) — MEDIUM, best "core AI + product" balance.

Description: User recites; app transcribes with a Quran-tuned Whisper, aligns to the canonical text (edit-distance/fuzzy match), highlights missed/wrong/skipped words, and schedules revision with spaced repetition.
Core AI: ASR (`tarteel-ai/whisper-base-ar-quran` or a small fine-tune) + alignment + SRS.
Data: EveryAyah/QuranicAudio for testing; Tanzil text.
Difficulty: Medium. *Differentiator vs. Tarteel:* scope to a limited surah set but add something Tarteel under-serves — **feedback quality for non-native/children's voices**, an **Urdu/regional-language feedback UI**, or transparent open evaluation of alignment accuracy.

4. Tajweed rule classifier for a small fixed rule-set — MEDIUM, strong academic core.

Description: Given a short recitation clip, detect correct vs. incorrect application of 3–5 tajweed rules (start from QDAT's Al Mad, Ghunnah, Ikhfaa; optionally add Qalqalah).
Core AI: CNN/CRNN/EfficientNet or LSTM on mel-spectrograms/MFCCs; librosa/Parselmouth features.
Data: **QDAT** (public, labeled); augment with your own recorded, expert-annotated clips.
Difficulty: Medium. *Differentiator:* reproduce and then **beat or extend** the published QDAT benchmarks (e.g., add rules, add explainable per-frame diagnosis, or test cross-reciter generalization) — a clean, publishable contribution and not a clone of any app.

5. Non-native / children's Quran pronunciation feedback (makharij) — MEDIUM-HARD, high-impact niche.

Description: Focus on the *articulation points* where non-Arabic speakers (very relevant in Pakistan) systematically err (e.g., س/ص, د/ض, ك/ق, ه/ح), giving targeted, letter-level feedback.

Core AI: wav2vec2/HuBERT phoneme-level mispronunciation detection & diagnosis (MDD), possibly phonological-feature CTC.
Data: You will likely need to **collect and annotate** a small dataset (a documented FYP deliverable in itself); seed with EveryAyah "correct" references.
Difficulty: Medium-hard (data scarcity is the crux). *Differentiator:* an explicitly **non-native, regional** focus — a documented gap, since most datasets are professional male Arab reciters.

6. Word-level "Shazam for the Quran" verse identifier — MEDIUM.

Description: Recite any fragment; the app identifies surah + ayah.
Core AI: ASR → fuzzy text search over the full Quran index.
Data: Quran text + any Quran-tuned ASR.
Difficulty: Medium. *Differentiator:* robustness evaluation across reciters/accents/noise, or optimizing for on-device/offline use.

7. Personalized adaptive hifz tutor combining ASR feedback + SRS + analytics — HARDER (integrative).

Description: Ideas #3 + a learner model that adapts revision to each user's *historical mistakes* (like Tarteel's adaptive review), with a dashboard.
Core AI: ASR + alignment + a recommendation/scheduling model.
Difficulty: Hard (multiple subsystems). *Differentiator:* the **adaptive learner-modeling** and its evaluation with real users.

8. Quranic camera OCR + reader for the visually-impaired or for annotation — MOST AMBITIOUS.

Description: Photograph a mushaf page → recognize verses (with diacritics) → read aloud / link to tafsir.
Core AI: CNN+RNN/Transformer OCR on Uthmani script.
Data: Mushaf al-Madinah OCR dataset, QTID.
Difficulty: Hard (diacritic OCR is unsolved at high accuracy). *Differentiator:* an accessibility angle (audio + translation for low-literacy or visually-impaired users).

Recommended pick for a 4-person team: Idea **#3 or #4** as the AI core, wrapped with Idea **#1 or #2** as a secondary feature — this gives you a genuine, evaluable AI contribution *and* a demoable product, while staying within a 6–9 month timeline.

4. Common challenges & pitfalls

Religious/scholarly accuracy is non-negotiable. Wrong tajweed feedback or a hallucinated ruling is worse than no app. Egypt's Dar al-Iftā and Indonesia's Nahdatul Ulama have warned against AI issuing fatwas; the consensus in the literature (IslamicLLM surveys, Aftina) is *source-grounding + citation verification + human scholar oversight + refusal on uncertainty.* **Recruit a qari/scholar advisor and put "reviewed by a scholar" in your methodology.**
Data scarcity for tajweed-labeled audio. Only QDAT (~1,505 clips, 3 rules, one verse) is truly open and labeled; most audio is *correct* professional recitation with no error labels. Budget time to record and expert-annotate your own small set, and scope your rule-list accordingly.

Demographic bias in training data. Professional-male-Arab reciters dominate; models fail on women, children, and non-native speakers. Either target this gap deliberately or acknowledge it as a limitation.
Arabic NLP tooling limitations. Diacritics carry meaning, morphology is rich, and Quranic Arabic differs from Modern Standard Arabic — generic Arabic tools underperform. Use Quran-specific resources (Quranic Arabic Corpus, CAMeL Tools, Quran-tuned Whisper).
Chatbot hallucination and doctrinal bias. LLMs conflate weak and sahih hadith, oversimplify fiqh, and can favor one school. Mitigate with strict retrieval, re-ranking, similarity thresholds, refusal, and clear "educational, not a fatwa" disclaimers. OpenReview ↗
Scope creep / cloning Tarteel. The classic failure mode is trying to rebuild a 15M-user product. Scope to a few surahs / a few rules / one language / one user group, and compete on *evaluation and a specific niche*, not breadth.
Evaluation rigor. Use proper metrics (WER for ASR, per-rule accuracy/F1 for tajweed, citation-accuracy/hallucination-rate for RAG), a held-out test set, and ideally release your benchmark — this is what makes it *research*.

Recommendations

Weeks 1–4 (proposal): Pick **one AI core** (recommended: tajweed classification on QDAT, or scoped recitation follow-along on Juz Amma) + **one secondary feature** (grounded RAG Q&A or semantic search). Write the literature review around the papers above (arXiv 2305.06429, 2503.23470, 2509.00094, 2311.05120; the Tarteel NeurIPS 2021 dataset paper). Line up a **qari/scholar advisor** now.
Weeks 5–8 (data + baseline): Stand up the data pipeline from open sources (Tanzil, Al Quran Cloud, EveryAyah, QDAT). Get a *baseline* working end-to-end (e.g., off-the-shelf `whisper-base-ar-quran` + fuzzy alignment, or an MFCC+SVM tajweed baseline) before any fine-tuning. **Benchmark to hit:** reproduce a published number (e.g., ~95% on a QDAT rule, or WER ≤10% on your test set).
Weeks 9–20 (build + improve): Fine-tune/upgrade the model, add the secondary feature, build the app UI.
Threshold to advance: if your tajweed classifier can't beat ~90% on your chosen rules, narrow to fewer rules or more data rather than adding features. If your RAG hallucinates on your scholar-reviewed test questions, tighten retrieval/refusal before adding a nicer UI.
Weeks 21–28 (evaluate + defend): Run rigorous evaluation, collect a small real-user study (great for defense), document limitations honestly, and **release your code + any new labeled data** on GitHub/Hugging Face (this strengthens the academic contribution and your CVs).
Decision triggers to *change course*: If data annotation stalls past week 8 → drop to a pure text/RAG project (no audio). If compute is a bottleneck → use `whisper-tiny`/LoRA or shift to the classification/RAG track. If the scholar advisor flags accuracy problems you can't fix → reframe the deliverable as "an assistant that defers to human scholars," which is itself a defensible, honest design.

Caveats

Some competitor claims are marketing, not verified. Tarteel's "15+ million" users (from its own store listing) and its community-reported ">96% error detection / NeMo Conformer-CTC" architecture are company- or third-party-reported, not independently verified primary sources; treat as indicative.

The frontier tajweed system (QPS, 0.16% PER) is a preprint (arXiv 2509.00094, ICML 2026 listing), trained largely on error-free "golden" recitation, so its real-world error-detection accuracy on genuine learner mistakes is not yet fully established — do not treat it as a settled baseline.
Dataset details vary by source (e.g., QDAT is cited as both ~1,505 and 1,509 clips; EveryAyah hours differ between studies). Verify exact figures against the primary papers when you cite them.
App names collide ("Noor AI"/"NoorAI"/"NurAI" are several unrelated products; multiple "Quran AI"/"AL Quran AI" apps exist) — confirm which specific app you mean before comparing.
License compliance matters: Tanzil requires attribution and a link and forbids altering the text; Quranic Arabic Corpus is GNU-licensed; the Tarteel dataset is CC BY-NC-ND 4.0 (non-commercial, no-derivatives); some datasets (e.g., Quran-Lab's ASR benchmark) forbid redistribution or TTS/voice-cloning use. Read the terms before publishing.